

PRESENTATION

SOFTWARE PROJECT LAB - I

Project Title : **The El do'raão Explorer**

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PROJECT OVERVIEW

El Dorado Explorer is a multi-modal local retrieval system designed to efficiently search and extract information from large-scale heterogeneous data sources.

RETRIEVAL STRATEGY

- Follows a two-phase retrieval approach.
- **Initial query:**
 - Uses direct brute-force processing.
 - Performs full computation (PDF parsing, pixel-level analysis).
- **Subsequent queries:**
 - Use module-specific cached representations.
 - Avoid repeated expensive computations.
 - Result in significantly faster execution.

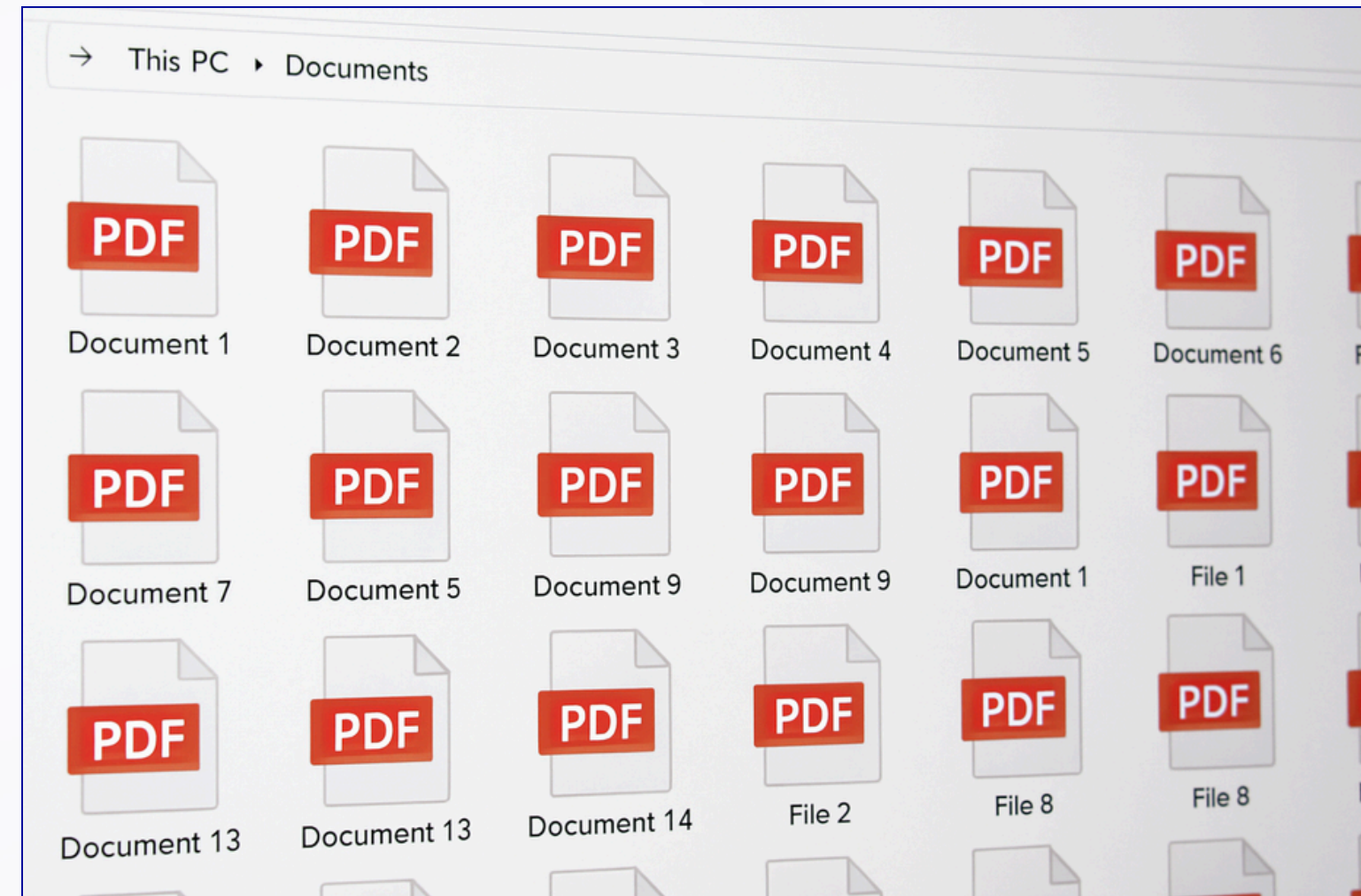
CURRENT MODULES

- Supports pluggable **search modules** (extensible design).
- **Current modules:**
 1. **PDF text retrieval** (large-scale document search).
 2. **Bayesian-based image skin segment retrieval.**

MODULE-1

MOTIVATION – 1

- Thousands of PDF files stored
- Need to find specific content quickly
- Manual search is inefficient and time-consuming



MODULE-1

PDF Text Retrieval

Performs keyword search over **PDFs** with linear scanning and cached text representations for efficient repeated retrieval.

MODULE-1

PDF Text Retrieval

```
ashik@ashik-HP:~/Desktop/Sem 3/Software-Project-Lab$ ./a.out
```

Select Mode Options -

- (1) PDF Content Search
 - (2) BMP Human Skin Search
- => 1

Select Search Scope Options -

- (1) Whole Computer
 - (2) Specify Roots
- => 2

Number of roots = 1

Enter roots =>

1 - Files

Select Search Depth Options -

- (1) With Sub Directories
 - (2) Without Sub Directories
- => 1

Inputs -

- Search Mode - PDF or Image
- Search Scope
- Number of Roots
- Specifying Root names
- Search Depth

MODULE-1

PDF Text Retrieval

```
Select Options -  
(1) Whole Word  
(2) Partial word  
=> 1  
  
Select Search Level Options -  
(1) Quick Search  
(2) Deep Search (With Suggestion)  
=> 1  
  
Select Keyword Mode Options -  
(1) Single Keyword Search  
(2) Multiple Keyword Search  
(3) Multiple Keyword in Single Page Search  
=> 1  
Enter keywords  
1 - Faith  
  
Select Case Sensitivity Options -  
(1) Case Sensitive  
(2) Case Insensitive  
=> 2
```

Inputs -

- Whole or partial word?
- Search Level
- Keyword Mode
- Specifying Keywords
- Case sensitivity

MODULE-1

PDF Text Retrieval

```
=> 2
Files/2.pdf
1) Page No - 1 :
- (30/37) beliefs). This dual focus enhances the academic understanding of
- (33/37) Coexistence, Mutual Respect and Understanding.

2) Page No - 2 :
- (41/45) pervasive misunderstandings. The explanation of the Sufism legacy
- (43/45) faith understanding, cross-cultural survival, and spiritual nourishment

3) Page No - 10 :
- (21/43) understanding.26

4) Page No - 16 :
- (15/42) Understanding the dual nature of Sufi influenceboth positive

5) Page No - 19 :
- (30/51) place of interfaith dialogue and understanding and ensuring Islams

6) Page No - 23 :
- (84/86) Understanding the formation of Silsila-e-Furfura Sharifin West Bengal.

Files/sub/2.pdf
7) Page No - 1 :
- (32/39) academic understanding of Sufisms complex legacy in the region.
- (35/39) Understanding.

8) Page No - 2 :
- (41/45) pervasive misunderstandings. The explanation of the Sufism legacy
- (43/45) faith understanding, cross-cultural survival, and spiritual nourishment

9) Page No - 10 :
- (21/43) understanding.26

10) Page No - 16 :
- (15/42) Understanding the dual nature of Sufi influenceboth positive

11) Page No - 19 :
- (23/44) place of interfaith dialogue and understanding and ensuring Islams

12) Page No - 23 :
- (58/60) Understanding the formation of Silsila-e-Furfura Sharifin West Bengal.
```

Output -

- File paths inside roots
- Page numbers
- Line numbers
- Target Lines with keywords
- Suggestion of words

MODULE-1

PDF Text Retrieval

```
Which one you want to open? (0 for exit)  
=> |
```

Navigation -

- User enters the serial number.
- The corresponding PDF opens at the specified **page** in the browser.

MODULE-1

PDF Text Retrieval

If **Deep Search** is ON, then it shows **WORD SUGGESTIONS**

Keyword : Religion

```
69) Page No - 19 :  
Suggestions :  
  *Levenshtein - reception, region, reign, relationship, religious,  
  *Jaro-winkler - region, reign, relationship, religious,  
  *Soundex -
```

Keyword : Present

```
7) Page No - 7 :  
Suggestions :  
  *Levenshtein -  
  *Jaro-winkler - poorest, professed,  
  *Soundex - preaching, preachings,
```

MODULE-1

PDF Text Retrieval

Three word-matching algorithms -

1. **Levenshtein distance**
2. **Jaro-winkler similarity**
3. **Soundex**

MODULE-1

PDF Text Retrieval

Problem - Efficiency ?

Text is repeatedly extracted from the same PDF files !!

PDF **text extraction** is time-consuming and computationally expensive !!

MODULE-1

PDF Text Retrieval

CACHE MECHANISM

- Stores extracted text from PDFs after the first search.
- Eliminates repeated PDF-to-text conversion.

```
home/  
├─ documents/  
│   ├── doc1.pdf  
│   ├── doc2.pdf  
│   └─ doc3.pdf  
└─ application/ (collapsed)
```

Original File Structure

```
home/  
├─ documents/ (collapsed)  
└─ application/  
    └─ cache/  
        └─ home/  
            └─ documents/  
                ├── doc1.txt  
                ├── doc2.txt  
                └─ doc3.txt
```

MIRROR File Structure

MODULE-1

PDF Text Retrieval

CACHE MECHANISM

```
cacheSample.txt X
cacheSample.txt
1  ---LAST MODIFICATION TIME---
2  -----NUMBER OF PAGES-----
3  ----- PAGE-1 CONTENT -----
4  ----- BLANK SPACE -----
5  ----- PAGE-2 CONTENT -----
6  ----- BLANK SPACE -----
7  .
8  .
9  .
10 EOF
```

Sample Cache File

MODULE-1

PDF Text Retrieval

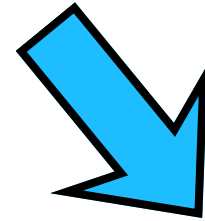
OVERALL CACHE WORKFLOW (HIGH-LEVEL VIEW)

```
PDF_SEARCH(){  
    - Iterate through PDF Files  
    - if ( PDF file does have a Cache File ) {  
        if(The Cache File is valid)  
            SEARCH_CONTENT(Cache File);  
        else{  
            DELETE_OLD_CACHE(PDF File);  
            SEARCH_CONTENT(Convert_to_text(PDF File));  
            STORE_CACHE(PDF File);  
        }  
    }else{  
        SEARCH_CONTENT(Convert_to_text(PDF File));  
        STORE_CACHE(PDF File);  
    }  
}
```

MODULE-1

PDF Text Retrieval

CACHE OUTPUT



```
Files/2.pdf (cached)
1) Page No - 7 :
- (22/46) steadfast dedication to the Islamic faith. This exhibition left Battuta

2) Page No - 16 :
- (41/42) receptive to a faith that went beyond material standards of wealth,

3) Page No - 17 :
- (4/42) faiths emphasis on values like equality, coexistence in a multicultural
- (19/42) the Islamic faith as a beacon of light and tried to lead their lives

4) Page No - 22 :
- (30/51) 7 Gokta_, Vahit, and Saeyd Rashed Hasan Chowdury. Freedom of Religion, Faith
```

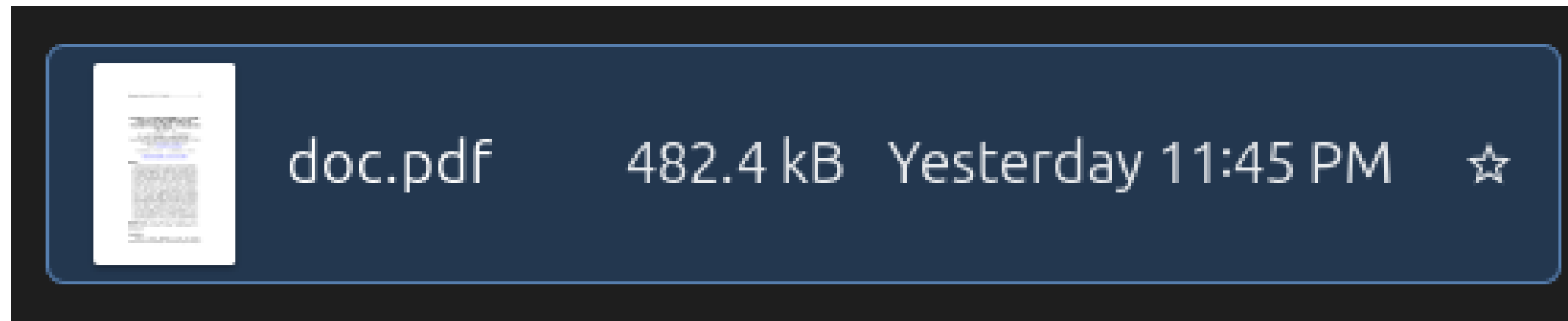
Here, **Search Result** is retrieved from the cache file.

MODULE-1

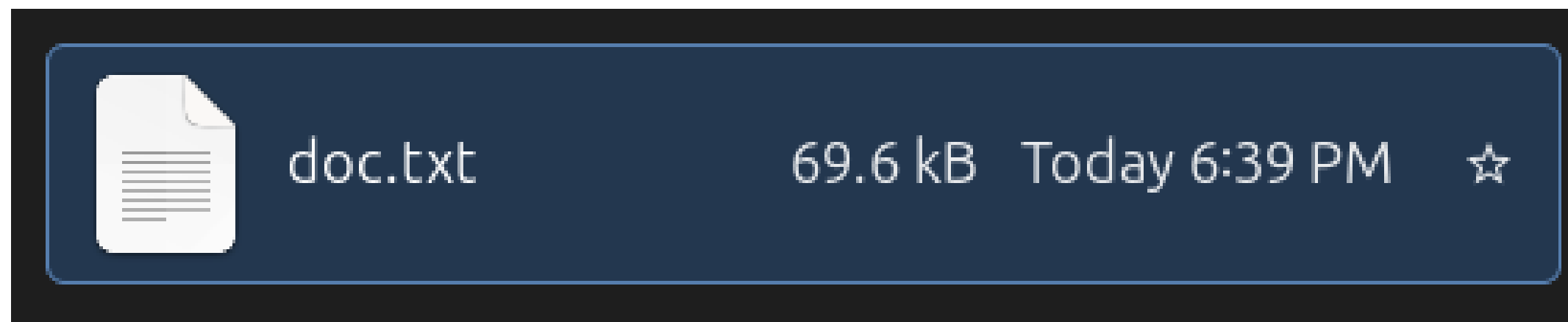
PDF Text Retrieval

CACHE EFFICIENCY

A normal PDF File size –

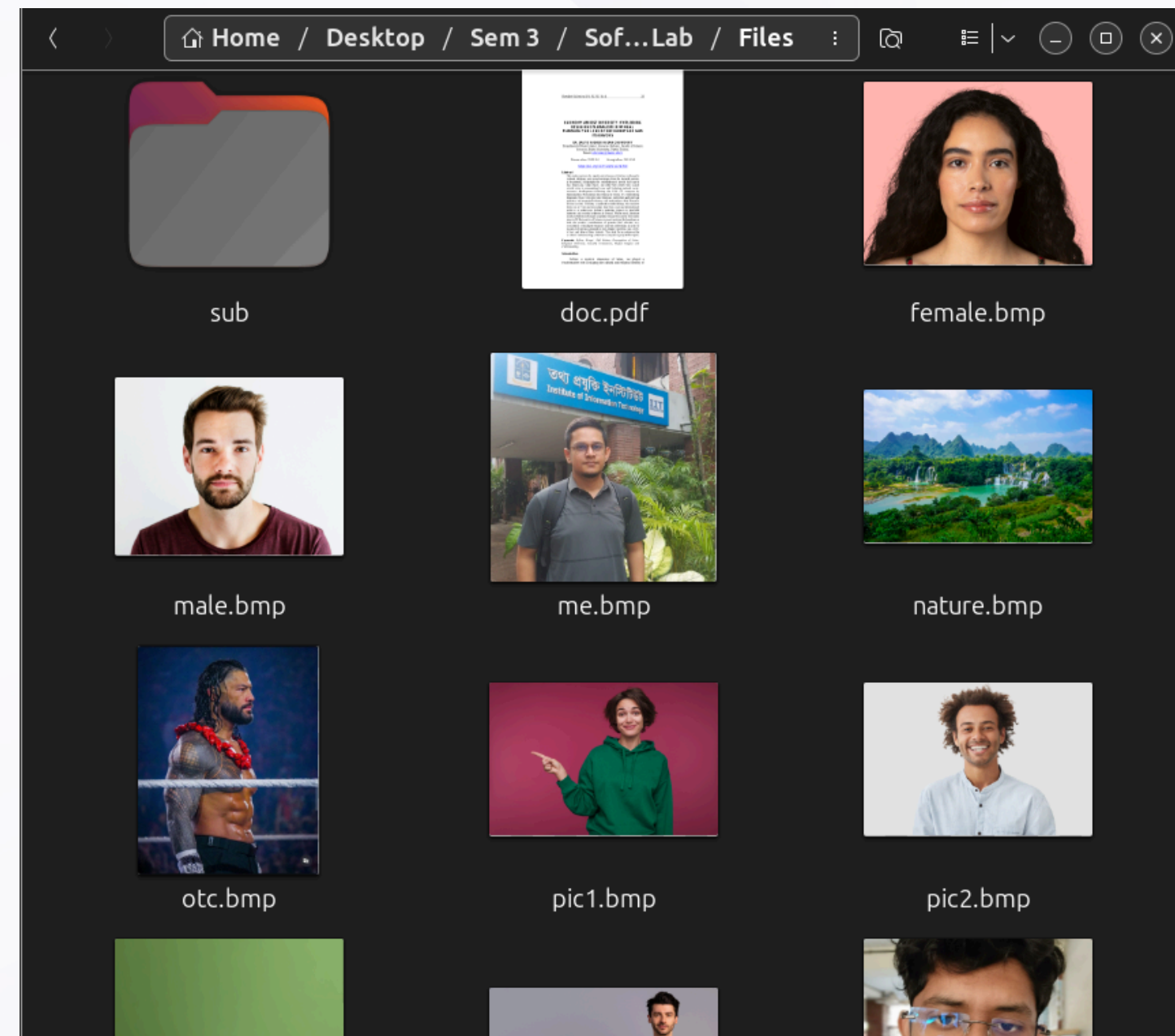


Its Cache File size –



MOTIVATION – 2

- Thousands of Image files stored.
- Need to search and detect **HUMAN SKIN** regions from the images.



MODULE-2

MODULE-2

Skin Region Retrieval

The system includes an **image retrieval module** that identifies and extracts **human skin regions** using a pixel-level **Naive Bayes classification model**.

MODULE-2

Skin Region Retrieval

```
Welcome to
THE El Do'raão EXPLORER

Select Mode Options -
(1) PDF Content Search
(2) BMP Human Skin Search
=> 2

Select Model Options -
(1) Model-1 (Good for family images)
(2) Model-2 (Good for single images)
=> 2
Enter root => Files

Select Recursivity Options -
(1) Without sub-directories
(2) With sub-directories
=> 1
Retrieving from cache
```

Inputs -

- Search Mode - PDF or Image
- Model Version
- Specifying Root name
- Search Depth

MODULE-2

Skin Region Retrieval

```
Predicting skin segment from the Image!
```

```
(1) Files/shapa.bmp  
(2) Files/male.bmp  
(3) Files/pic3.bmp  
(4) Files/nature.bmp  
(5) Files/me.bmp  
(6) Files/pic2.bmp  
(7) Files/pic4.bmp  
(8) Files/rokib.bmp  
(9) Files/otc.bmp  
(10) Files/female.bmp  
(11) Files/iit-day.bmp  
(12) Files/06Apr03Face.bmp  
(13) Files/pic1.bmp
```

```
Which one do you want to open? (0 to exit)
```

```
=> █
```

Outputs -

- Performs **pixel-level** human skin region **classification**.
- Generates and caches **skin-segmented image representations**
- Allows the user to select which image's skin-segmented result to view.

MODULE-2

Skin Region Retrieval

CACHE MECHANISM

- Stores precomputed skin-segmentation results.
- Avoids repeated pixel-level classification.

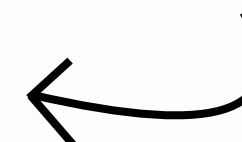
Original File Structure

```
└─ home/  
  └─ pictures/  
    └─ pic1.bmp  
    └─ pic2.bmp  
    └─ pic3.bmp  
  └─ application/ (collapsed)
```

MIRROR File Structure

```
└─ home/  
  └─ pictures/ (collapsed)  
  └─ application/  
    └─ cache/  
      └─ home/  
        └─ pictures/  
          └─ 48372615_pic1.bmp  
          └─ 94720384_pic2.bmp  
          └─ 60149277_pic3.bmp
```

Last Modification Time



MODULE-2

Skin Region Retrieval



Original Image



Predicted Skin Segment
by Model

MODULE-2

Skin Region Retrieval



Original Image



Ground Truth



Predicted by Model

DEPENDENCIES

OS - Linux

Language - C++

Integrated Development Environment (IDE) - VS code

Version Control System - Git / Github

Library - Poppler

Dataset: Human Skin Dataset that will be used to train and test skin detection across diverse skin tones and lighting conditions.



THANK YOU